

Student Name: Gaurav V R

Register Number: 715523106032

Institution: PSG Institute of Technology and Applied Research

Department: Electronics and Communication Engineering

Date of Submission: 25/04/2025

1. Problem Statement

News spreads quickly via the internet, and research suggests that increasing numbers of adults rely on social media for their news. The purpose of some sites is fun and entertainment, but the intentions of others are insidious. Some readers read stories with outright lies to laugh, knowing they are fake, while others readers are less wary. Thus, it has become more prominent to check the authenticity of the news. Currently, the news is weighed for its truth based on self-awareness. But there comes a point where too much information is loaded, which makes it difficult to validate it all manually. Thus, we chose the problem statement, “Exposing the truth with advanced fake news detection powered by natural language processing” to tackle the same problem.

2. Objectives of the Project

- ❖ The phenomenon of Fake news is experiencing rapid and growing progress with the evolution of the means of communication and Social media. Fake news detection is an emerging research area which is gaining big interest. It faces some challenges due to the limited resources such as datasets and processing and analysing techniques. In this work, we propose a system for Fake news detection that uses machine learning techniques.
- ❖ The goal of fake news detection is to develop algorithms that can automatically identify and flag fake news articles, which can be used to combat misinformation and promote the dissemination of accurate information.
- ❖ As a consequence of the above, fake news has become one of the major concerns

because of its potential to destabilize governments, which makes them a potential danger to modern society. An example of this can be found in the US. electoral campaign, where the term “fake news” gained great notoriety due to the influence of the hoaxes in the final result of these.

- ❖ Our ultimate goal is to build an AI model that processes the raw news lines and uses NLP techniques to train and test the model to predict fake news in real time.

3. Scope of the Project

Features to build:

1. **Text classification:** A machine learning model to classify the news as fake or real and label them with multiple classes
2. **Stylometric analysis:** To understand the writing style, grammar and emotional tone. This can indicate typical inconsistencies found in fake news
3. **Cross-Checking system:** Fake news can be compared with factual claims. Knowledge graphs can be used to verify the originality of information. This can be used to analyse the information alongside source reputation scoring
4. **Behavioral analysis:** We get valid pieces of information by observing how the information flows in social media

Limitations and Constraints

- Inadequate large, labeled datasets
- Varying performance of the model due to underfitting/overfitting to biases or weights
- Combining text and media is difficult
- Ethical and data privacy issues

Tools and libraries

1. Scikit-Learn

2. Tensorflow/Pytorch
3. OpenCV
4. NLTK/Spacy

Data Sources

The data source which will probably be used for this project is LIAR dataset which contains 3 files with .tsv format for test, train and validation. Below is some description about the data files used for this project.

LIAR: A BENCHMARK DATASET FOR FAKE NEWS DETECTION

William Yang Wang, "Liar, Liar Pants on Fire": A New Benchmark Dataset for Fake News Detection, to appear in Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (ACL 2017), short paper, Vancouver, BC, Canada, July 30-August 4, ACL.

It is a public, static dataset which contains 13 variables/columns for train, test and validation sets. Other optional datasets are also being searched for better flexibility and efficiency.

NELA: A LARGE MULTI-LABELLED NEWS DATASET FOR MISINFORMATION IN NEWS ARTICLES

NELA-GT-2020 contains nearly 1.8M news articles from 519 sources. These sources come from a wide range of mainstream news sources and alternative news sources. Included in the dataset are source-level ground truth labels from Media Bias/Fact Check (MBFC) covering multiple dimensions of veracity.

4. High-Level Methodology

● Data Collection:

Marketing scholars increasingly use web scraping and application programming interfaces (APIs) to collect data from the internet. Yet, despite the widespread use of such web data, the idiosyncratic and sometimes insidious challenges in its collection have received limited attention. While existing resources emphasize technical details of

extracting web data, the authors propose a novel methodological framework focused on enhancing its validity. In particular, the framework highlights how addressing validity concerns requires the joint consideration of idiosyncratic technical and legal/ethical questions along the three stages of collecting web data: selecting data sources, designing the data collection, and extracting the data. Reviews of more than 300 articles using web data were published in the top five marketing journals and offer a typology of how web data have advanced marketing thought.

- **Data Cleaning:**

Data cleaning and preparation is an essential process in any data analysis project, particularly when working with bond data. Bond data may be complicated, heterogeneous, and erroneous and inconsistent as a result of several factors including data sources, formats, standards, and conventions.

Handling missing values:

Missing values are a prevalent issue in bond data since some fields of data may not be reported or available for a particular bond or period. Missing values can impact the validity and adequacy of our analysis and introduce uncertainty or bias in our findings. We hence need to spot and handle missing values suitably.

Deleting or ignoring :

This is the easiest and the most direct one to handle missing values, but it might not be the best choice in most situations. Removing or excluding missing values can decrease the size and representativeness of our data set, and could also introduce bias or distortion in our analysis if the missing values are not random or independent of other variables. Thus, we can only utilize this procedure if there are scarce and non-substantive missing values and if they would not influence the validity or generalizability of our research.

Imputing or filling:

This is the technique of substituting or approximating the missing values with some methods or rules, e.g., mean, median, mode, interpolation, regression, or machine learning. Imputing or filling missing values can assist us in maintaining the size and shape of our data set, and can also enhance the accuracy and stability of our analysis. Yet, we must be vigilant and honest about the limitations and assumptions of our imputation or filling approach, and we must also validate and test the quality and effect of our filled or imputed values on our analysis. For instance, we can employ cross-validation, sensitivity analysis, or comparison with alternative data sources to test the validity and reliability of our filled and imputed value.

Flagging or marking:

It is the operation of marking or flagging the missing values with certain symbols or codes, e.g., NA, NULL, or -999. Flagging or marking missing values can enable us to trace the original position and state of missing values and also enable us to use different methods or techniques to deal with them based on the context and intention of our analysis. For instance, we can utilize marking or flagging to omit or include missing values in our aggregations, calculations, or visualization, or use various imputation or filling techniques on various forms of missing values.

Handling outliers:

Outliers are values that are way too distant from other data points, as there might be error in one measurement or another due to presence of some errors of entry, natural variation etc. Outliers may bias our distribution and also our summary statistics about our data along with our performance and results of techniques that we perform on our data like regression, clustering, and classification. We have to handle and identify outliers subsequently. Some potential methods for how we could treat outliers are:

1. Detection or identification: It is the process of finding or identifying the outliers in our data using some techniques or criteria, for instance, box plots, histograms, scatter plots, z-scores, or interquartile ranges. Identifying or detecting outliers can help us know the patterns and properties of our data, and may also find some unusual or interesting findings or anomalies in our data. However, we should note that the definition and identification of outliers may be context-dependent and domain-dependent to our data, and that outliers may be valid or meaningful data points that reflect the complexity or reality of our data. Therefore, we ought not to delete or convert outliers blindly or mechanically without closer examination or explanation.

2.Deleting or filtering: Removing or excluding the outliers from our data or analysis manually or programmatically. Outliers may be filtered or deleted to reduce the noise and variability of our data, and to increase the accuracy and stability of our methods of analysis. Nevertheless, we must be cautious and careful about the impact and implications of removing or censoring outliers because this can affect our data set size and representativeness and our analysis being potentially biased or skewed if the outliers are not randomly distributed or are independent of the other variables. Therefore, we can apply this method only if the outliers are clearly incorrect or irrelevant and do not threaten the validity or the generalizability of our analysis.

3.Rescaling or adjusting: This entails the practice of changing or modifying the values or scales of the outliers by some functions or techniques, such as winsorizing, trimming, capping, or log-transforming. Reprocessing or modifying outliers may enable us to normalize or standardize our data, as well as optimize the performance and result of

our analysis methods, especially outlier-stricken or sensitive procedures such as linear regression, correlation, or mean. However, we must be cautious and strict about assumptions and limitations of our adjusting or transforming processes, and we must document and disclose adjustments and implications of our altered or transformed values on our analysis. For example, we may use descriptive statistics, plots, or tables to illustrate the distribution and summary of our raw and adjusted or transformed values, and we may also use metrics, tests, or models to compare the performance and results of our analysis before and after adjusting or transforming outliers

- **Exploratory Data Analysis (EDA):**

Exploratory Data Analysis plays a crucial role in uncovering data patterns and relationships. We understand the characteristics of data behaviour, point out potential issues, and gain useful insights that can guide feature engineering.

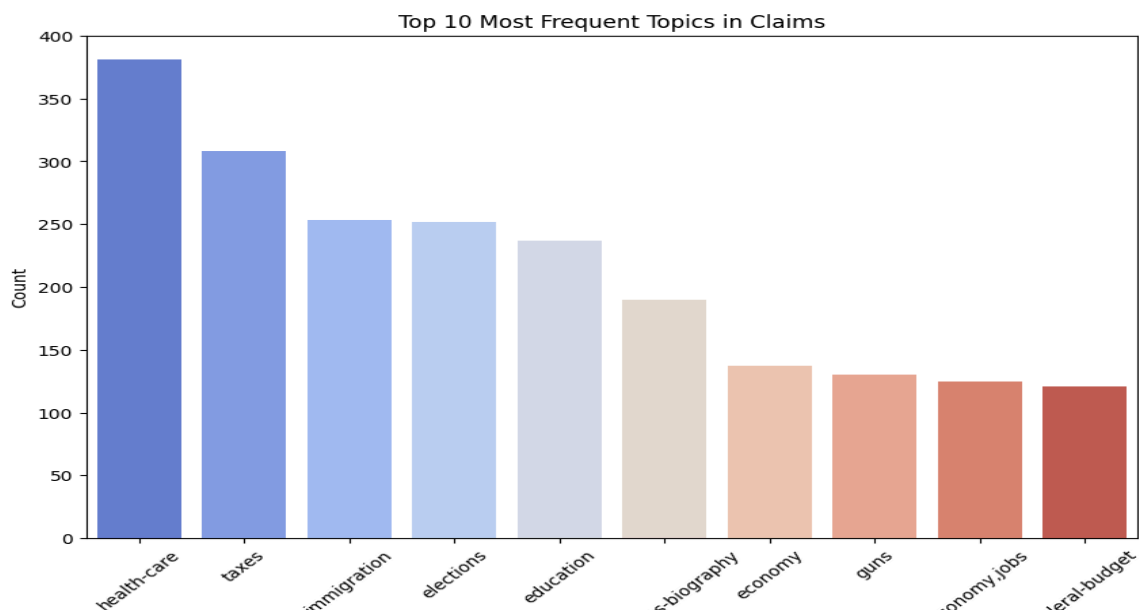
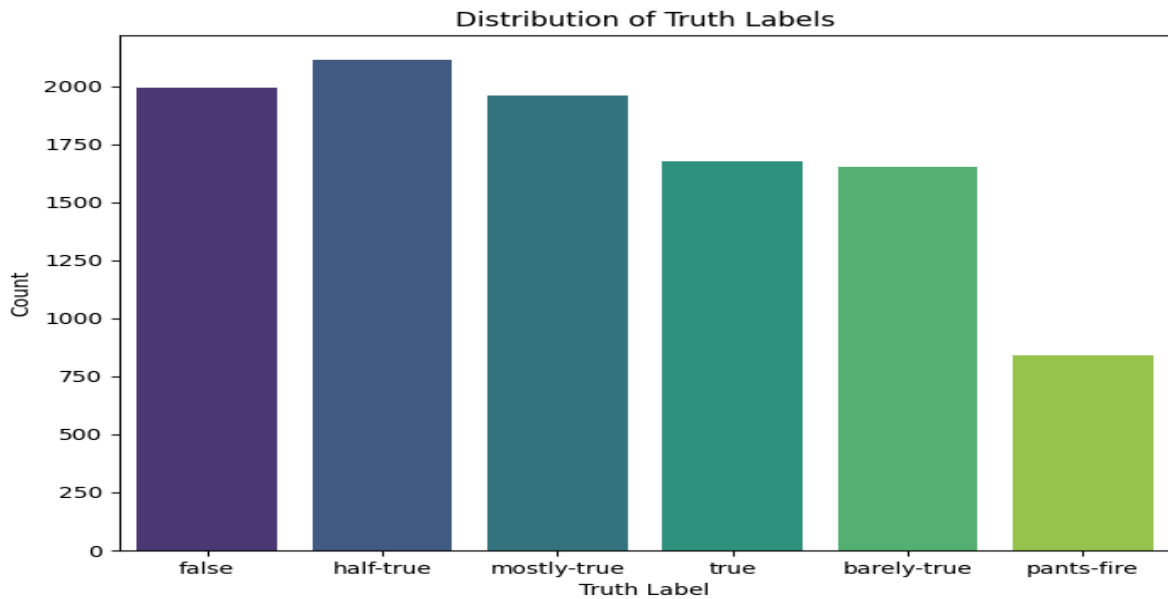
Data Distribution: Examining the data distribution involves classification tasks, since we classify the news as fake vs. real. This can be viewed as standard bar graphs

Text Data Analysis: NLP models capture frequently occurring terms in the dataset, by which common words that help in identifying the keywords are examined to differentiate fake news from real news. The TF-IDF(Term Frequency-Inverse Document Frequency) metric helps in identifying words that are particularly important in the document's content compared to their frequency across the entire dataset. Bar plot helps in understanding this context better

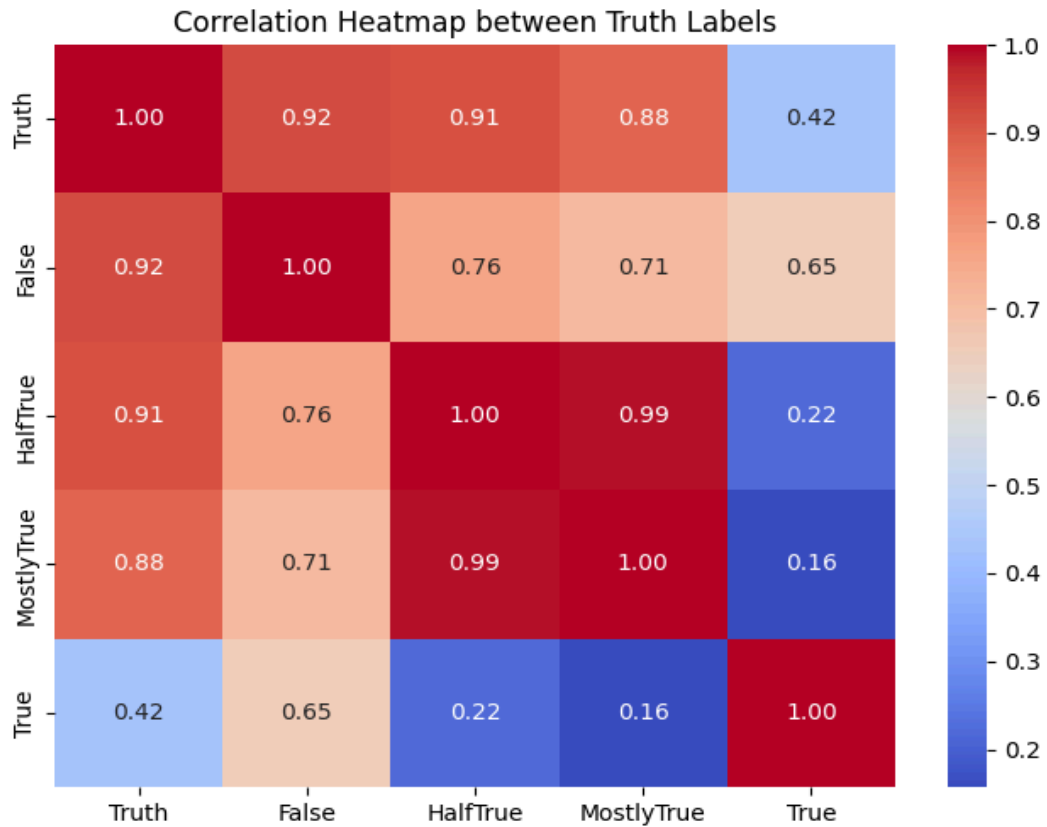
Sentiment Analysis: Fake news may show different/unique sentiment compared to real news. Degree of sentiment and subjectivity of the

information gives better information, which can be visualized using scatter plots

The graphs below are data visualization of Truth-labels and Claims for different cases in LIAR dataset:



Correlation of numerical columns- Truth,False,Half-true,Mostly-true,Truth



Correlation heatmaps: With this we could visualize strong relationships between numerical features, which may aid in classification process

Topic Modeling: Latent Dirichlet Allocation(LDA) can uncover hidden information within the news articles

Time-Series Plots: These plots show the count of fake and real news articles overtime to observe trends and seasonal patterns

● Feature Engineering

In machine learning, the performance of a model relies heavily on the quality and relevance of the features that are used for training. Feature engineering is critical in finding significant information from raw data and generating new features that enhance the model's predictive ability. This technique entails feature transformation, feature combination, and feature selection to enhance the model's accuracy, generalizability, and resilience. In this blog, we will understand why feature engineering is important and show its real-world application through code examples using Python and leading machine-learning libraries.

Feature engineering is the process of transforming raw data into useful features that simplify the machine learning algorithm learning process. A proper set of features can significantly influence model performance, and ill-engineered features can lead to overfitting or underfitting. Based on the data's domain knowledge and problem understanding, feature engineering enables us to extract useful patterns and relationships and improve model accuracy and interpretability.

Raw data itself might not be immediately consumable by machine learning algorithms. Raw data is usually needed to be preprocessed, and feature engineering serves as a bridge between raw data and valuable insights. Feature engineering is where the domain knowledge enters in deciding what are good features that capture hidden patterns in the data.

Categorical variables are non-numerical variables representing various categories or groups. Numerical inputs are usually required by machine learning algorithms, and thus categorical variables have to be mapped into an appropriate numerical format. Feature engineering offers techniques such as one-hot encoding, label encoding, and target encoding to convert categorical data to numerical representations in an

efficient manner.

One-Hot Encoding: In one-hot encoding, a category is represented as a binary vector, and the corresponding element is 1 for the category and 0 for other categories. For instance, if we have a "Gender" feature with categories "Male" and "Female," one-hot encoding will generate two binary features: "Gender_Male" and "Gender_Female."

Label Encoding: In label encoding, a category is represented by an integer value. For instance, "Male" could be represented as 0 and "Female" as 1. Label encoding is appropriate for ordinal categorical variables where the categories have a certain order.

Target Encoding: Target encoding involves substituting every category with the mean (or some other summary statistic) of the target variable for that category. Target encoding can be handy when working with high-cardinality categorical variables.

6.Tools and Technologies

- **Programming Language** – *Python, HTML, CSS, Oracle SQL, Java*
- **Notebook/IDE** – *Jupyter Notebook*
- **Libraries** – *Pandas, Numpy, scipy, scikit-learn, etc.*

7.Team Members and Roles

Tt Members	Roles
Dousik Manokaran	Design and Test Engineer
Gaurav V R	Data Engineer
Dillimaran K	ML Specialist and Project Manager